

LDOCX ENTERPRISE SPECIFICATION

LIVING DOCUMENT FORMAT: ARCHITECTURE, SECURITY & 3D ENGINEERING GUIDE

The definitive technical manual for Senior Developers, Security Engineers, and 3D Artists to inspect, extend, and master the LDOCX ecosystem.

TARGET AUDIENCE

Senior Developers, AppSec, 3D Artists

SPEC VERSION

2.4.0 (Reactive 3D Runtime)

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I. EXECUTIVE SUMMARY & PARADIGM SHIFT

The **LDOCX (Living Document XML/eXtended)** format represents a generational leap in digital publishing and presentation technology. While traditional formats (such as PDF and static EPUB) treat documents as flat, unreactive print emulations, **LDOCX reimagines documents as self-contained, intelligent, reactive 3D software runtimes.**

```
+-----+ | LDOCX LIVING DOCUMENT | | |
+-----+ +-----+ +-----+ | | | Tactile Paper Feel | | Interactive
| | In-Engine 4K | | | 3D Corner Lift & | | WebGL 3D Models | | Fluid Water & | | | Real Paper Physics
| | & HoLo Shaders | | Particle Waves | | +-----+ +-----+
--+ | | | +-----+ +-----+ +-----+ | | | Universal In-Canvas |
| Dynamic Forms | | Embedded Local | | | Free Text Writing | | & Stripe Pay | | Reactive State | | | &
Direct Typography | | Checkout Links | | & AI Assistant | | | +-----+ +-----+
+ +-----+ +-----+ +-----+ +-----+
```

Capability	Traditional PDF	Web Slides (PPTX/Keynote)	LDOCX Living Document
Interactivity	Static vector/raster	Linear button links	True 60 FPS WebGL, Shaders, Sandboxes
Tactile Feel	Rigid flat page	Slide transitions	Real-time 3D paper flex & hand pressure physics
Free-Form Text	Requires Acrobat Pro	Text box locks	Direct in-canvas double-click free writing everywhere
Media Coupling	Passive video embed	Simple play/pause	Keystrokes trigger real-time water wave ripples
Data Capture	Flat AcroForms	External links	Direct Stripe payment gateways & webhook forms
Portability	Single file	Presentation file + media	Zero-dependency .ldocx archive (ZIP container)

2. SECTION 1: THE SENIOR DEVELOPER’S ARCHITECTURAL GUIDE

2.1 File Format Anatomy (.ldocx)

An `.ldocx` package is an uncompressed or deflate-compressed ZIP container containing structured JSON metadata, reactive AST page trees, and media assets.

```
document.ldocx (ZIP Archive) |-- manifest.json ← Package metadata, schema version, page index |-- pages/ |
|-- page_1.json ← AST node tree for Slide 1 | |-- page_2.json ← AST node tree for Slide 2 | `--
page_n.json |-- assets/ | |-- models/ ← GLTF/GLB/OBJ binary models | |-- textures/ ← Environment maps,
bump/roughness maps | |-- audio/ ← Ambient atmospheric sound loops (MP3/OGG) | `-- video/ ← Moving
photograph cutouts (MP4/WebM) |-- scripts/ ← Custom sandbox logic (optional) `-- checksum.sha256 ←
Package integrity verification hash
```

2.2 Abstract Syntax Tree (AST) Specification

```
interface LDocNode {
  type: 'heading' | 'paragraph' | 'blockquote' | 'list' | 'list_item' | 'table'
    | 'image' | 'web_image' | 'video' | 'web_video' | 'audio'
    | 'model_3d' | 'jsx_canvas' | 'particles' | 'water_effect'
    | 'button' | 'form' | 'preorder';
  value?: string;
  level?: number;
  style?: {
    src?: string;
    action?: 'link' | 'navigate' | 'submit' | 'save_state';
    target?: string;
    btn_style?: 'royal_gold' | 'neon_cyan' | 'purple_nebula' | 'minimal';
    code?: string;
    mode?: 'water' | 'stardust' | 'warp' | 'embers';
    model_format?: 'gltf' | 'glb' | 'obj';
    model_material?: 'cyber_hologram' | 'royal_gold' | 'obsidian_carbon' | 'crystal_cobalt';
    checkout_url?: string;
    webhook_url?: string;
  };
  children?: LDocNode[];
}
```

2.3 Multi-Style Presentation Artifact Engine

- 📜 **Enchanted Tactile Paper:** Warm fibrous parchment gradient with realistic 3-tier elevation drop shadows and letterpress ink bleed text.
- 💎 **Frosted Glass Holographic Cards:** Translucent frosted glassmorphism with dynamic 3D cursor tilt and iridescent specular highlights.
- 🎬 **Cinematic Keynote Deck:** 16:9 widescreen layout with high-impact typography and dark obsidian background.

- 📰 **Living Editorial Gazette**: Dual-column magazine/newspaper layout with decorative drop-caps.

2.4 Real-Time Tactile Paper Physics Engine

```
function attachTactilePaperPhysics(page) {
  page.addEventListener('mousemove', (e) => {
    if (currentPresentationMode !== 'paper') return;
    const rect = page.getBoundingClientRect();
    const px = ((e.clientX - rect.left) / rect.width) - 0.5;
    const py = ((e.clientY - rect.top) / rect.height) - 0.5;

    // Realistic human hand 3D tilt & dynamic elevation shadow
    const rx = -py * 7;
    const ry = px * 7;
    const shadowX = -px * 38;
    const shadowY = Math.max(16, 26 + py * 26);
    const shadowBlur = 48 + Math.abs(px * 28);

    page.style.setProperty('--paper-rx', rx.toFixed(2) + 'deg');
    page.style.setProperty('--paper-ry', ry.toFixed(2) + 'deg');
    page.style.setProperty('--paper-shadow',
      shadowX.toFixed(0) + 'px ' + shadowY.toFixed(0) + 'px ' + shadowBlur.toFixed(0) + 'px');
    page.classList.add('paper-lifted');
  });
}
```

2.8 EXTENSION PACK v2.5.0: UNIVERSAL EDITOR SUITE & POP-UP SHELL

The v2.5.0 architecture introduces additive modules for complete living document authoring versatility:

- **Module E (Unified Pop-Up Shell):** Draggable, non-modal frosted glass container (`backdrop-filter: blur(28px)` , iridescent borders). Enforces that only *one heavy panel* (Photo Studio or 3D Scene Editor) is active at a time to preserve 60 FPS.
- **Module A (Rich Document Editor – Docx Mode):** Formatted text nodes (`rich_text`), advanced tables (`table_pro`), and review comments (`comment_thread`). Floating Format Popover HUD appears on selection.
- **Module C (Photo Studio):** Non-destructive `image_layer_stack` descriptors on image nodes (crop, 90 deg rotation, filters, brush overlay). Original assets in `assets/` are never overwritten.
- **Module B (PDF / Acrobat-Style Layer):** `annotation_layer` and `form_field` nodes (highlighter, sticky notes, freehand pen, signature pad) with `POST /api/export/pdf` flattening.
- **Module D (3D Scene Editor):** Unity-style 3-pane overlay (Hierarchy, Viewport with Three.js WebGL & Gizmos, Inspector with the 4 PBR presets). Real-time polygon budget monitor (<35,000 tris).

3. SECTION 2: THE SECURITY ENGINEER’S HARDENING MANUAL

3.1 Threat Model & Attack Vectors

Vector	Threat	Mitigation in LDOCX
ZIP Archive Ingestion	Zip Slip / Directory Traversal (escaping via <code>../</code>)	Strict filepath canonicalization and prefix validation before decompression.
Embedded Sandboxes	DOM XSS, Token Theft, Cookie Access	Isolated <code><iframe sandbox="allow-scripts"></code> omitting <code>allow-same-origin</code> .
User-Generated Content	Cross-Site Scripting via inline HTML tags	Strict <code>escapeHtml()</code> applied across all text rendering routines.
Local REST API Server	LAN exposure, Remote Execution	Engine binds strictly to <code>127.0.0.1</code> (loopback interface only).
Rich-Text Paste	DOM XSS via clipboard HTML (Word / Docs)	Strict allow-list tag & attribute sanitizer before <code>escapeHtml()</code> ; never raw <code>innerHTML</code> .
Image / 3D Upload	Decompression bomb, oversized texture, malformed GLTF	Enforce size ceiling (max 10MB texture) and polygon budget (<35k tris) in AST parser & server.
POST /api/export/pdf	Unauthorized export / loopback bypass	Strict Origin and Referer validation enforcing <code>127.0.0.1</code> loopback identity.

3.2 Safe ZIP Extraction Verification

```
func sanitizeZipPath(targetDir, filePath string) (string, error) {
    cleanPath := filepath.Clean(filePath)
    dest := filepath.Join(targetDir, cleanPath)
    if !strings.HasPrefix(dest, filepath.Clean(targetDir) + string(filepath.Separator)) {
        return "", fmt.Errorf("security violation: illegal path traversal: %s", filePath)
    }
    return dest, nil
}
```

4. SECTION 3: THE 3D ARTIST & CREATIVE TECHNOLOGIST BLUEPRINT

4.1 Three.js WebGL Engine Integration

```
ThreeJsViewer Instance |-- Scene (THREE.Scene) |-- Camera (THREE.PerspectiveCamera, 45 deg FOV) |--
WebGLRenderer (antialias: true, alpha: true) |-- Lighting Rig | |-- AmbientLight (0xffffffff, intensity 0.6)
| |-- DirectionalLight Key (0xf59e0b, intensity 1.2, top-right) | `-- DirectionalLight Fill (0x38bdf8,
intensity 0.8, bottom-left) `-- Render Loop (60 FPS with motion throttling window.__ldocLivingMotion)
```

4.2 PBR Material Presets

Material ID	Description	Key Shading Parameters
royal_gold	Polished 24K Gilded Gold	color: 0xf59e0b, metalness: 0.94, roughness: 0.16, envMapIntensity: 1.5
cyber_hologram	Sci-Fi Wireframe Hologram	color: 0x38bdf8, emissive: 0x0284c7, wireframe: true, opacity: 0.85
obsidian_carbon	High-Gloss Carbon Weave	color: 0x111118, metalness: 0.3, roughness: 0.7
crystal_cobalt	Optical Refractive Glass	color: 0x60a5fa, transmission: 0.9, roughness: 0.1, ior: 1.52

4.3 3D Artist Asset Export Checklist

- File Format:** Export binary .glb with Draco compression enabled.
- Polygon Count:** Keep triangle count under 35,000 tris per model for optimal 60 FPS mobile/desktop rendering.
- Textures:** PBR metallic-roughness workflow; pack Occlusion, Roughness, and Metallic into an ORM texture.
- Pivot & Scale:** Center the model pivot at (0, 0, 0) and normalize scale to fit within a 2-unit bounding sphere.

✓ Document Sign-Off

This specification provides everything necessary for Senior Full-Stack Engineers, Application Security Professionals, and 3D Artists to build, audit, and create living documents within the LDOCX ecosystem.